

Robust Controller Synthesis

Thijs Lenssen and Amritam Das

Abstract—

*Index Terms—*PDEs, Robust Control, LMIs;

I. INTRODUCTION

This paper uses PN PI operators inspired by the works [2], [7], [13]. Then it IQC extends the results from [1], [4], [11]. [3] also covers robust control using dual iq

STRUCTURE OF THE PAPER

The structure of the paper will be organized as follows:

- **Preliminaries:** This section introduces the necessary notation, concepts related to PI and PIETOOLS, and the IQC result that will be central to the subsequent developments in the paper.
- **Scaled IO Result:** In this section, we define the scalings associated with both the primal and dual systems. These scalings play a critical role in the connection between the primal and dual formulations. Then we will use these scalings in a scaled IO duality result.
- **Connection to IQC:** We then connect the scaled duality result to IQC. Consider the following hard IQC condition:

$$\int_0^T \left\langle \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \Pi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt \geq 0$$

where the dynamic multiplier $\Pi(j\omega)$ can be decomposed into a dynamic part $\Psi(j\omega)$ and a static part $\mathcal{V}$. The IQC condition then becomes:

$$\int_0^T \left\langle \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \mathcal{V} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt \geq 0$$

We propose a further decomposition of $\mathcal{V}$ as:

$$\mathcal{V} = \mathcal{S}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \mathcal{S}$$

where $\mathcal{S}$ represents the scaling operations applied to the output of $\Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}$. Assuming Ψ admits a PIE system, we augment the nominal system as $z_\Delta = Gw_\Delta$. Rewriting the IQC condition with this decomposition gives:

$$\int_0^T \left\langle \mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt$$

This formulation reduces to analyzing the open-loop system $\mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}$. After establishing the necessary

assumptions and conditions, we apply the scaled duality result to the augmented system with corresponding scalings.

- **Controller Synthesis:** In this section, we propose Linear PI Inequalities (LPIs) for robustly synthesizing controller gains. If time allows, we will extend this to include output feedback. Additionally, we may explore using the DK iteration method to solve for suboptimal solutions when combining dynamic multipliers with output feedback.
- **Examples and Validation:** Finally, we explore practical applications, focusing on performance shaping using weighting filters. The results will be validated through Frequency Response Function (FRF) measurements.

II. PRELIMINARIES

A. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. For a normed space X and Banach space Y , $\mathcal{L}(X, Y)$ denotes the Banach space of bounded linear operators from X to Y with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and $\mathcal{L}(X) := \mathcal{L}(X, X)$. Our notational convention is to write functions in **bold** (e.g., $\mathbf{x}$, indicating $\mathbf{x} \in \mathbb{R}L_2$) and operators in calligraphic font (e.g., $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$). For a Hilbert space X and operator $\mathcal{A} \in \mathcal{L}(X)$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^* \mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space $L_2^n[0, \infty)$ denotes $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain $[0, \infty)$, so that $x(t) \in \mathbb{R}^n$. Similarly, $\mathbb{R}L_2^n[0, \infty)$ denotes $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$, we write $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$. For a Hilbert space $\mathbf{H}$, we denote by $\mathbf{L}_\mathbf{H}$ the space of square-integrable functions $u : [0, \infty) \rightarrow \mathbf{H}$ with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_\mathbf{H}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

T. Lenssen and A. Das are with the Control Systems Group, Eindhoven University of Technology, The Netherlands. Email addresses: t.m.lenssen@student.tue.nl, am.das@tue.nl
Corresponding author: am.das@tue.nl

We define the extended space $\mathbf{L}_{e,\mathbf{H}}$ to consist of functions $u : [0, \infty) \rightarrow \mathbf{H}$ such that

$$\int_0^T \|u(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use $\mathbf{L}_{e,[a,b]}$ and $\mathbf{L}_{[a,b]}$ to denote $\mathbf{L}_{e,\mathbf{H}}$ and $\mathbf{L}_{\mathbf{H}}$ with $\mathbf{H} = \mathbb{R}L_2[a, b]$. For any Hilbert space $\mathbf{H}$, the truncation operator $p_\tau : \mathbf{L}_{e,\mathbf{H}} \rightarrow \mathbf{L}_{e,\mathbf{H}}$ is defined for $\mathbf{y} \in \mathbf{L}_{e,\mathbf{H}}$ by

$$(p_\tau \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

such that $(p_\tau \mathbf{y})(t) \in \mathbf{L}_{\mathbf{H}}$ for all $\tau \geq 0$.

Definition 1. Let $\mathbf{H}$ and $\mathbf{G}$ be Hilbert spaces. For all $\tau \geq 0$, an operator $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$ is:

- **Causal** if $p_\tau G = p_\tau G p_\tau$.
- **Bounded on $\mathbf{L}_{\mathbf{H}}$** if $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$.
- **Bounded on $\mathbf{L}_{e,\mathbf{H}}$** if $\|p_\tau G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_\tau \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{e,\mathbf{H}}$.

Finally, for a given $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, we define the operator $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e,[a,b]}^n \rightarrow \mathbf{L}_{e,[a,b]}^m$ to compactly represent a multiplication operator. For $\mathbf{u} = [u_1 \cdots u_N]^\top$ with $\mathbf{u} \in \mathbb{R}L_2$, its action at time t is given by the block structure of $\mathcal{K}_{ij}$:

$$(\mathcal{M}_{\mathcal{K}}\mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t). \quad (1)$$

B. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space.

Definition 2. We say $\mathcal{P} \in \Pi_4^{m \times n} \subset \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, denoted by $\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix}$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\left[\left(\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix} \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R}\mathbf{x})(s) = R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta.$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and concatenation – implying that Π_4 is a composition algebra [10].

Definition 3. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\mathcal{L}(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\begin{aligned} & \min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\}) \\ & \text{s.t.} \quad \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \begin{bmatrix} P_i & Q_{1i} \\ Q_{2i} & \{R_i\} \end{bmatrix} \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0 \end{aligned} \quad (2)$$

Only LMIs are required to solve an optimization problem involving LPIs. These definitions are based on [8]

Definition 4 (PIE System). Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{n_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2[a, b]^{n_x}$ and $z(t) \in \mathbb{R}L_2^{n_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \quad (3)$$

for all $t \geq 0$ and $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$. Additionally, for signals $\underline{w}(t) \in \mathbb{R}L_2[a, b]^{n_z}$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2[a, b]^{n_x}$ and $\underline{z}(t) \in L_2[a, b]^{n_w}$ satisfy the dual PIE, $G^\top$ with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix}, \quad (4)$$

for all $t \geq 0$ and $\mathcal{T}^* \underline{\mathbf{x}}(0) = \mathcal{T}^* \underline{\mathbf{x}}_0$.

Definition 5 (Generalization of $\mathbb{RH}_\infty$). Let G be a PIE system as in Definition 4. We say G is internally stable and bounded with decay rate $\alpha > 0$, $M > 0$, and L_2 -gain $\gamma > 0$ if

- The homogeneous part $\partial_t(\mathcal{T}\mathbf{x})(t) = \mathcal{A}\mathbf{x}(t)$, with $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$ is exponentially stable such that, for every admissible initial condition $\mathbf{x}_0 \in \mathbb{R}L_2$

$$\|\mathcal{T}\mathbf{x}(t)\|_{\mathbf{L}_{[a,b]}} \leq M\|\mathcal{T}\mathbf{x}_0\|_{\mathbf{L}_{[a,b]}} e^{-\alpha t}, \quad \forall t \geq 0.$$

- With zero initial condition $\mathbf{x}(0) = 0$, then for any input w and output z ,

$$\|p_\tau z\|_{\mathbf{L}_{[a,b]}} \leq \gamma \|p_\tau w\|_{\mathbf{L}_{[a,b]}}, \quad \forall \tau \geq 0, \quad \forall.$$

Clearly, $\mathbb{RH}_\infty \subset \mathbb{RH}_\infty^{\text{PIE}}$ [12]. Also $G^\top$ is in $\mathbb{RH}_\infty^{\text{PIE}}$ with the same decay rate, L_2 -gain, and M [8].

C. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [9]. For learning how to use PIETOOLS and its functionalities visit PIETOOLS homepage.

III. IQC-BASED FRAMEWORK FOR ROBUST STABILITY AND PERFORMANCE ANALYSIS

This section develops a comprehensive framework for analyzing stability and performance of interconnected systems in a Hilbert space, utilizing IQCs as the primary analytical tool. We begin by establishing the mathematical foundation for system interconnections and formalizing the key concepts of stability and performance. The IQC methodology is then systematically developed to characterize system uncertainties

and derive rigorous conditions that guarantee robust stability and performance. A central contribution of this work lies in bridging IQC theory with μ -analysis for infinite dimensional systems, creating a framework for quantifying their robust stability and performance when those systems are subjected to structured uncertainties.

A. Description of uncertain systems in a Hilbert-space

We start by denoting the uncertainty operators as Δ that belongs to a structured set Δ which satisfies the following fundamental property:

Assumption 1. $\Delta \in \Delta$ implies that $\tau\Delta \in \Delta$ for all $\tau \in [0, 1]$.

That is, the uncertainty set is closed under scaling by factors in $[0, 1]$, which is the standard homotopy assumption used in IQC theory. We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator G and Δ . This interconnection structure is formally defined as follows

Definition 6. Given operators $G_\Delta : \mathbf{L}_{e,[a,b]}^{n_{w\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z\Delta}}$, $G_{\Delta w} : \mathbf{L}_{e,[a,b]}^{n_w} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z\Delta}}$, $G_{z\Delta} : \mathbf{L}_{e,[a,b]}^{n_{w\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_z}$, $G_{zw} : \mathbf{L}_{e,[a,b]}^{n_{w\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_z}$, and $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w\Delta}}$, we define an **interconnected system** $z = [G, \Delta](w)$, for inputs $w \in \mathbf{L}_{e,[a,b]}^{n_w}$ and outputs $z \in \mathbf{L}_{e,[a,b]}^{n_z}$, such that the signals $z_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{z\Delta}}$, $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w\Delta}}$ satisfy the following relations

$$\begin{aligned} z_\Delta &= G_\Delta w_\Delta + G_{\Delta w} w, \\ z &= G_{z\Delta} w_\Delta + G_{zw} w, \\ w_\Delta &= \Delta z_\Delta. \end{aligned} \quad (5)$$

The interconnection structure described above captures the feedback interaction between the nominal system G and the uncertainty Δ , where w and z represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 7. We say that the uncertain interconnection $[G, \Delta]$ is **well-posed** if:

- **Existence and Uniqueness:** For any $w \in \mathbf{L}_{e,[a,b]}$, there exist unique $z \in \mathbf{L}_{e,[a,b]}$, $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}$ such that, z , w , z_Δ , w_Δ satisfy the interconnection.
- **Causality:** If z_Δ , w_Δ and $\hat{z}_\Delta$, $\hat{w}_\Delta$ satisfy the interconnection, then $p_\tau(z_\Delta - \hat{z}_\Delta) = 0$ and $p_\tau(w_\Delta - \hat{w}_\Delta) = 0$, for all $\tau \geq 0$.

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 6.

B. Robust stability and performance

Traditional control system design often treats stability and performance shaping as separate objectives with distinct constraints. However, Integral Quadratic Constraints (IQCs)

provide a unified framework where these trade-offs can be captured within a single constraint [11]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (5), and introducing auxiliary signals z_0 and w_0 .

Definition 8. The interconnection

$$z_\Delta = G_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$, is said to be **robustly stable** if:

- 1) The uncertain system is well-posed for all $\Delta \in \Delta$.
- 2) The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $\mathbf{L}_{[a,b]}$ for all $\Delta \in \Delta$.

Within the IQC framework, the stability of an interconnection $[G, \Delta]$ is certified by verifying that the uncertainty operator Δ satisfies a quadratic constraint defined by an appropriate multiplier. This leads to the following formal definition

Definition 9. An uncertainty operator $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w\Delta}}$ is said to satisfy the IQC defined by the operator $\Psi \in \mathbf{L}_{e,[a,b]}^{n_{w\Delta} + n_{z\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w\Delta} + n_{z\Delta}}$ multiplier $\mathcal{M}_\Psi \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_\Delta = \mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ if the inequality holds for all $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0. \quad (6)$$

Moreover, let $\mathcal{V}_\Delta \{ \mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta \}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 10. Robust performance is achieved if the interconnection $[G, \Delta]$ is **robustly stable** for some performance multiplier $\mathcal{M}_{\mathcal{V}_p} \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ with $\mathcal{Q}_p \succeq 0$ such that, for all $w \in \mathbf{L}_{e,[a,b]}$, all $T \geq 0$, and some sufficiently small $\epsilon > 0$,

$$\left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

Moreover, let $\mathcal{V}_p \{ \mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p \}$ be parameterized as,

$$\mathcal{V}_p = \begin{bmatrix} \mathcal{Q}_p & \mathcal{S}_p \\ \mathcal{S}_p^* & \mathcal{R}_p \end{bmatrix}$$

In this framework, the multipliers $\mathcal{M}_\Psi$ and $\mathcal{M}_{\hat{\Psi}}$ serve distinct but complementary roles. The multiplier $\mathcal{M}_\Psi$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the multiplier $\mathcal{M}_{\hat{\Psi}}$ encodes performance

requirements for the nominal system, such as L_2 -gain bounds, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives.

C. Theorem for joint robust stability and performance

We now present the main theoretical result that provides conditions for simultaneous robust stability and performance verification.

Theorem 1 (Robust Stability and Performance). *Let $G, \Delta \in \Delta$ be given, and there exists $\Psi, \mathcal{M}_{V_\Delta}$, such that,*

- *The interconnection $[G, \Delta]$ is well-posed for all $\Delta \in \Delta$*
- *The IQC in Definition 9 is satisfied for all $\Delta \in \Delta$.*

*Then the interconnection $[G, \Delta]$ is **robustly stable** and **robust performance** is achieved with respect to some $\mathcal{M}_{V_p}$, in Definition 10, with $\mathcal{Q}_p \succeq 0$, if,*

$$\left\langle p_T \hat{G} \begin{bmatrix} w_\Delta \\ w \end{bmatrix}, p_T \begin{bmatrix} \mathcal{M}_{V_\Delta} & 0 \\ 0 & \mathcal{M}_{V_p} \end{bmatrix} \hat{G} \begin{bmatrix} w_\Delta \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w\|_{\mathbf{L}_{[a,b]}}^2) \quad (7)$$

for all $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$, $w \in \mathbf{L}_{e,[a,b]}^{n_w}$ and $T \geq 0$, where $\hat{G} := \begin{bmatrix} \Psi & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} G_\Delta & G_{\Delta w} \\ I & 0 \\ -G_{z\Delta} & -G_{zw} \\ 0 & I \end{bmatrix}$ according to (5).

Proof. See Appendix A.1 ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

IV. DUALITY UNDER DYNAMICAL FILTERING

Theorem 2 (Dual L_2 -gain [8, Thm. 9]). *For a PIE system with corresponding PI operators $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$, the following statements are equivalent.*

- 1) *For any $w \in \mathbf{L}_{e,[a,b]}^{n_w}$, and $\mathbf{x}(0) = 0$, G satisfies $\|P_\tau z\| \leq \|P_\tau w\|$.*
- 2) *For any $\underline{w} \in \mathbf{L}_{e,[a,b]}^{n_z}$, and $\underline{\mathbf{x}}(0) = 0$, $G^\top$ satisfies $\|P_\tau \underline{z}\| \leq \|P_\tau \underline{w}\|$.*

Definition 11 (Primal and Dual Dynamic Scalings). *Let $\Psi = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix}$ satisfy $\Psi, \Psi^{-1} \in \mathbb{RH}_\infty^{\text{PIE}}$. Let $\hat{\mathcal{J}} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. The dual filter associated with Ψ is defined by the dual operation*

$$\mathbf{D}(\Psi) := (\hat{\mathcal{J}}^* \Psi^\top \hat{\mathcal{J}})^{-1} = \begin{bmatrix} \Psi_w^\top & -\Psi_z^\top \\ -\Psi_{wz}^\top & \Psi_z^\top \end{bmatrix}^{-1}. \quad (8)$$

Note $\mathbf{D}(\Psi) \in \mathbb{RH}_\infty^{\text{PIE}}$. Since $\hat{\mathcal{J}}^ = \hat{\mathcal{J}}^{-1}$, $\hat{\mathcal{J}}^* = -\hat{\mathcal{J}}$, $(\Psi^\top)^\top = \Psi$, and $(\Psi^{-1})^{-1} = \Psi$, this operation is an involution:*

$$\mathbf{D}(\mathbf{D}(\Psi)) = \Psi. \quad (9)$$

The filter Ψ admits the state-space realization

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi^z & \mathcal{B}_\Psi^w \\ \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z & \mathcal{D}_\Psi^{zw} \\ \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz} & \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ z(t) \\ w(t) \end{bmatrix}, \quad (10)$$

where the entries are PI operators of appropriate dimension. The dual filter $\mathbf{D}(\Psi)$ is equivalently represented through the inverse relation $\mathbf{D}(\Psi)^{-1} = \hat{\mathcal{J}}^ \Psi^\top \hat{\mathcal{J}}$, which has the state-space realization*

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\mathbf{x}}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi^* & -\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*} \\ -\mathcal{B}_\Psi^{w*} & \mathcal{D}_\Psi^{w*} & -\mathcal{D}_\Psi^{zw*} \\ \mathcal{B}_\Psi^{z*} & -\mathcal{D}_\Psi^{wz*} & \mathcal{D}_\Psi^{z*} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix}. \quad (11)$$

The filtered primal system is obtained from

$$\Psi \begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix}. \quad (12)$$

The filtered dual system is obtained from

$$\mathbf{D}(\Psi) \begin{bmatrix} \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}. \quad (13)$$

Adopted from [13]

Definition 12. (Potapov–Ginzburg Transform) *Let $\Psi : \begin{bmatrix} z \\ w \end{bmatrix} \rightarrow \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}$, where $\Psi = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix}$. Then the Potapov–Ginzburg transform of Ψ is defined as*

$$\begin{bmatrix} \tilde{z} \\ w \end{bmatrix} = \Psi^\dagger \begin{bmatrix} \tilde{w} \\ z \end{bmatrix}, \text{ where } \Psi^\dagger = \begin{bmatrix} \Psi_{zw} \Psi_w^{-1} & \Psi_z - \Psi_{zw} \Psi_w^{-1} \Psi_{wz} \\ \Psi_w^{-1} & -\Psi_w^{-1} \Psi_{wz} \end{bmatrix}. \quad (14)$$

Assumed to admit the state-space representation

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \hat{\mathcal{A}}_\Psi & \hat{\mathcal{B}}_\Psi^{\tilde{w}} & \hat{\mathcal{B}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^z & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} & \hat{\mathcal{D}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} & \hat{\mathcal{D}}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \\ z(t) \end{bmatrix}, \quad (15)$$

where

$$\begin{aligned} \hat{\mathcal{A}}_\Psi &:= \mathcal{A}_\Psi - \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, & \hat{\mathcal{B}}_\Psi^{\tilde{w}} &:= \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1}, \\ \hat{\mathcal{B}}_\Psi^z &:= \mathcal{B}_\Psi^z - \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{C}}_\Psi^z &:= \mathcal{C}_\Psi^z - \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, \\ \hat{\mathcal{D}}_\Psi^z &:= \mathcal{D}_\Psi^z - \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} &:= \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1}, \\ \hat{\mathcal{C}}_\Psi^w &:= -(\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} &:= (\mathcal{D}_\Psi^w)^{-1}, \\ \hat{\mathcal{D}}_\Psi^{wz} &:= -(\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}. \end{aligned}$$

Lemma 1. (Potapov–Ginzburg Transform Equivalence) *The Scaled Primal System (respectively, Scaled Dual System) is equivalent, under an invertible congruence transformation, to the Primal System (respectively, Dual System) augmented with the Potapov–Ginzburg transform of the Primal Scaling (respectively, Dual Scaling).*

Proof. We first show the primal derivation explicitly. To show sufficiency, first augment the **Primal System** with the Potapov–Ginzburg transform of the **Primal Scaling**. Substitution of $z = \mathcal{C}\mathbf{x} + \mathcal{D}w$ into (15) gives the algebraic loop

$$w = \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C}\mathbf{x} + \Lambda_p \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Collecting these blocks and retaining $\tilde{w}$ as an output yields

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} - \mathcal{B}\Lambda_c \mathcal{D}_\Psi^{wz} \mathcal{C} & -\mathcal{B}\Lambda_c \mathcal{C}_\Psi^z & 0 & 0 \\ \mathcal{B}_\Psi^z \mathcal{C} - (\mathcal{B}_\Psi^z \mathcal{D} + \mathcal{B}_\Psi^w) \Lambda_c \mathcal{D}_\Psi^{wz} \mathcal{C} & \mathcal{A}_\Psi - (\mathcal{B}_\Psi^z \mathcal{D} + \mathcal{B}_\Psi^w) \Lambda_c \mathcal{C}_\Psi^z & \mathcal{C}_\Psi^z - (\mathcal{D}_\Psi^z \mathcal{D} + \mathcal{D}_\Psi^w) \Lambda_c \mathcal{D}_\Psi^{wz} \mathcal{C} & 0 \\ \mathcal{D}_\Psi^z \mathcal{C} - (\mathcal{D}_\Psi^z \mathcal{D} + \mathcal{D}_\Psi^w) \Lambda_c \mathcal{D}_\Psi^{wz} \mathcal{C} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \\ \tilde{w}(t) \end{bmatrix} \quad (22)$$

The corresponding invertible congruence transformation is

$$\begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_\Psi^{wz} \mathcal{C} & \mathcal{C}_\Psi^w & \Lambda_c^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} \quad (23)$$

Applying (23) to (22) gives

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & 0 & \mathcal{B} \\ \mathcal{B}_\Psi^z \mathcal{C} & \mathcal{A}_\Psi & \mathcal{B}_\Psi^z \mathcal{D} + \mathcal{B}_\Psi^w \\ \mathcal{D}_\Psi^z \mathcal{C} & \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z \mathcal{D} + \mathcal{D}_\Psi^w \\ \mathcal{D}_\Psi^{wz} \mathcal{C} & \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz} \mathcal{D} + \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} \quad (24)$$

The realization in (24) is the direct augmentation of the **Primal System** with the **Primal Scaling**, and hence is the **Scaled Primal System**. For necessity, applying (23) to (24) recovers (22), and reversing (17)–(21) recovers (16). The dual case follows by applying the same argument to the **Dual System** and the **Dual Scaling**. ■

Theorem 3 (Scaled IO duality). *Assume the following conditions hold*

- The **Primal I/O Scaling** is a PI operator and invertible.
- $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ and $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ are invertible.

The following statements are equivalent.

- 1) For all $\tilde{w} \in \mathbb{RL}_2^{\mathbf{n}_w}$, the **Scaled Primal System** satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$.
- 2) For all $\tilde{\tilde{w}} \in \mathbb{RL}_2^{\mathbf{n}_z}$, the **Scaled Dual System** satisfies $\|P_\tau \tilde{\tilde{z}}\| \leq \|P_\tau \tilde{\tilde{w}}\|$.

Proof. See Appendix A.2. ■

V. DUAL IQCS

We also define the combined uncertainty and performance multiplier by

$$(\mathcal{M}_\mathcal{V} \mathbf{u})(t) := \underbrace{\begin{bmatrix} \mathcal{Q}_\Delta & 0 & \mathcal{S}_\Delta & 0 \\ 0 & \mathcal{Q}_p & 0 & \mathcal{S}_p \\ \mathcal{S}_\Delta^* & 0 & \mathcal{R}_\Delta & 0 \\ 0 & \mathcal{S}_p^* & 0 & \mathcal{R}_p \end{bmatrix}}_{\mathcal{V}} \mathbf{u}(t).$$

We restrict attention to multiplier operators that are compatible with the duality framework, and in particular to J -spectral operators for the static parts of the uncertainty and performance multipliers. Although J -spectral operators are usually introduced in frequency-domain settings, here they are used in the time domain to convert the IQC into a **Primal I/O Scaling** and a **Dual I/O Scaling**. This yields the relationship between the **Scaled Primal System** and **Scaled Dual System** described in Section IV. The defining factorization is

$$\mathcal{V} = \mathcal{V}^* = \mathcal{S}^{-*} \mathcal{J} \mathcal{S}^{-1} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix}, \quad \mathcal{J} = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix},$$

where $\mathcal{S}$ is the **Primal I/O Scaling**. This factorization may be interpreted as an input–output scaling of the primal and dual systems, converting the quadratic constraint into an equivalent open-loop representation. The associated dual J -spectral operator is

$$\bar{\mathcal{V}} := \bar{\mathcal{S}}^{-*} \mathcal{J} \bar{\mathcal{S}}^{-1} = \begin{bmatrix} -\mathcal{R}^* & \mathcal{S}^* \\ \mathcal{S} & -\mathcal{Q}^* \end{bmatrix}^{-1} = \begin{bmatrix} \bar{\mathcal{Q}} & \bar{\mathcal{S}} \\ \bar{\mathcal{S}}^* & \bar{\mathcal{R}} \end{bmatrix},$$

where $\bar{\mathcal{S}}$ is the **Dual I/O Scaling**. Many performance multipliers, as well as the static parts of stability multipliers, admit a J -spectral factorization. An important example is the class of strict PN-PI operators, adapted from [2].

Definition 13. Let $\mathcal{V} = \mathcal{V}^* \in \Pi_4$ be partitioned as

$$\mathcal{V} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix},$$

Then $\mathcal{V}$ is called a **strict PN-PI operator** if there exists $\epsilon > 0$ such that

$$\mathcal{Q} \succeq \epsilon I, \quad \mathcal{R} \preceq -\epsilon I.$$

If $\epsilon = 0$, then $\mathcal{V}$ is called a **PN-PI operator**.

Lemma 2. Let $\mathcal{V} = \mathcal{V}^* \in \Pi_4$ be a strict PN-PI operator. Then $\mathcal{V}$ admits the factorization

$$\mathcal{V} = \begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ 0 & \mathcal{S}_{22} \end{bmatrix}^{-*} \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ 0 & \mathcal{S}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathcal{S}_{11}^* \mathcal{S}_{11} & \mathcal{S}_{11}^* \mathcal{S}_{12} \\ \mathcal{S}_{12}^* \mathcal{S}_{11} & \mathcal{S}_{12}^* \mathcal{S}_{12} - \mathcal{S}_{22}^* \mathcal{S}_{22} \end{bmatrix} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix},$$

with $\mathcal{S}_{11} \succ 0$ and $\mathcal{S}_{22} \succ 0$.

Proof. See Appendix A.3. ■

Lemma 3. Let $\mathcal{V} = \mathcal{V}^* \in \Pi_4$ and $\bar{\mathcal{V}} = \bar{\mathcal{V}}^* \in \Pi_4$ denote the static primal and dual IQC multipliers. Then $\mathcal{V}$ is a strict PN-PI operator if and only if $\bar{\mathcal{V}}$ is a strict PN-PI operator.

Proof. See Appendix A.4. ■

VI. APPLICATION OF DUAL IQC

In this section, we will consider one application of the IQC duality result, namely robust controller gain synthesis, i.e. $u(t) = \mathcal{K} \mathbf{x}_f(t)$. We will represent the corresponding PIE system in the generalized plant format.

The IQC in (6) can be written as

$$\left\langle p_T \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0,$$

where $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} := v := \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is the output of the linear system Ψ driven by z_Δ and w_Δ with zero initial conditions. Let Ψ admit the realization

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi^{z_\Delta} & \mathcal{B}_\Psi^{w_\Delta} \\ \mathcal{C}_{\Psi,1} & \mathcal{D}_{\Psi,1}^{z_\Delta} & \mathcal{D}_{\Psi,1}^{w_\Delta} \\ \mathcal{C}_{\Psi,2} & \mathcal{D}_{\Psi,2}^{z_\Delta} & \mathcal{D}_{\Psi,2}^{w_\Delta} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}. \quad (25)$$

For the basis $\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix}$ function we require Ψ_{22} invertible, such that the Potapov–Grinsburg transform of Ψ is

well-defined. Typically, for the dynamic multipliers it results in the form $\Psi = \begin{bmatrix} \Psi_v & \Psi_v \\ \Psi_v & \Psi_v \end{bmatrix}$, i.e. if a partition is populated every partition has the same dynamics. Typically Ψ_v is taken tall such that it can represent the spectral landscape of the IQC well. We resolve this issue by 'squaring up' the dynamic

multiplier. If $\Psi_v = \begin{bmatrix} F_0(s) \\ F_1(s) \\ \vdots \\ F_v(s) \end{bmatrix}$, then this can be squared up by taking $\Psi_\square = \begin{bmatrix} F_0(s) & 0 \\ F_{1 \rightarrow v}(s) & I \end{bmatrix}$, this can be inverted if

$F_0(s)$ is invertible, then $\Psi_\square^{-1} = \begin{bmatrix} F_0^{-1}(s) & 0 \\ -F_{1 \rightarrow v}(s)F_0(s)^{-1} & I \end{bmatrix}$. This squaring up also introduces a lifted statespace.

Definition 14. [Generalized Plant] Let the PI operators be $\mathcal{T}, \mathcal{A} : \mathcal{L}(\mathbb{R}L_2^{n_x})$, $\mathcal{B}_\Delta : \mathbb{R}L_2^{n_{w\Delta}} \rightarrow \mathbb{R}L_2^{n_x}$, $\mathcal{B}_w : \mathbb{R}^{n_w} \rightarrow \mathbb{R}L_2^{n_x}$, $\mathcal{C}_\Delta : \mathbb{R}L_2^{n_x} \rightarrow \mathbb{R}L_2^{n_{z\Delta}}$, $\mathcal{D}_\Delta : \mathbb{R}^{n_{w\Delta}} \rightarrow \mathbb{R}L_2^{n_{z\Delta}}$, $\mathcal{D}_{\Delta w} : \mathbb{R}^{n_w} \rightarrow \mathbb{R}L_2^{n_{z\Delta}}$, $\mathcal{C}_z : \mathbb{R}L_2^{n_x} \rightarrow \mathbb{R}^{n_z}$, $\mathcal{D}_{z\Delta} : \mathbb{R}L_2^{n_{w\Delta}} \rightarrow \mathbb{R}^{n_z}$, $\mathcal{D}_{zw} \in \mathbb{R}^{n_z \times n_w}$, with $\mathbf{x}_f \in \mathbb{R}L_2^{n_x}$, regulated output $z \in L_{2,e}^{n_z}[0, \infty)$, disturbance $w \in L_2^{n_w}[0, \infty)$, and uncertainty $\Delta \in \Delta$, defined according to Definition 6, construct the following uncertain PIE,

$$\begin{aligned} \mathcal{T}\dot{\mathbf{x}}_f(t) &= \tilde{\mathcal{A}}(\delta)\mathbf{x}_f(t) + \tilde{\mathcal{B}}_w(\delta)w(t), \\ z(t) &= \tilde{\mathcal{C}}_z(\delta)\mathbf{x}_f(t) + \tilde{\mathcal{D}}_{zw}(\delta)w(t). \end{aligned} \quad (26)$$

The uncertain PIE (26) is then defined to admit a $[G, \Delta]$ form according to Definition 6 if there exist PI operators $\{\mathcal{T}, \mathcal{A}, \mathcal{B}_\Delta, \mathcal{B}_w, \mathcal{C}_\Delta, \mathcal{D}_\Delta, \mathcal{D}_{\Delta w}, \mathcal{C}_z, \mathcal{D}_{z\Delta}, \mathcal{D}_{zw}\}$ that parametrize the following PIEs for all $(\mathbf{x}_f(t), w_\Delta(t), z_\Delta(t), w(t), z(t))$, and for all $\Delta \in \Delta$,

$$G : \begin{cases} \mathcal{T}\dot{\mathbf{x}}_f(t) = \mathcal{A}\mathbf{x}_f(t) + \mathcal{B}_\Delta w_\Delta(t) + \mathcal{B}_w w(t), \\ z_\Delta(t) = \mathcal{C}_\Delta \mathbf{x}_f(t) + \mathcal{D}_\Delta w_\Delta(t) + \mathcal{D}_{\Delta w} w(t), \\ z(t) = \mathcal{C}_z \mathbf{x}_f(t) + \mathcal{D}_{z\Delta} w_\Delta(t) + \mathcal{D}_{zw} w(t), \end{cases}$$

with, $w_\Delta(t) = (\Delta z_\Delta)(t)$.

This definition is adapted from Corollary 8 in [11]

Definition 15 (Primal and Dual Augmented Systems). **Primal System G:** The primal system G is defined by the following equations: The augmented realization uses the partitioned filter output $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$. We assume there is a bounded selection operator $\mathcal{N}$ such that $w_\Delta(t) = \mathcal{N}v_2(t)$. The augmented input-output map is then written from $\begin{bmatrix} v_2^* & w^* \end{bmatrix}^*$ to $\begin{bmatrix} v_1^* & z^* \end{bmatrix}^*$.

$$\begin{aligned} \begin{bmatrix} \mathcal{T} & 0 \\ 0 & \mathcal{T}_\Psi \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}}_f(t) \\ \dot{\mathbf{x}}_\Psi(t) \end{bmatrix} &= \begin{bmatrix} \mathcal{A} & 0 \\ \mathcal{B}_\Psi^{z\Delta} \mathcal{C}_\Delta & \mathcal{A}_\Psi \end{bmatrix} \begin{bmatrix} \mathbf{x}_f(t) \\ \mathbf{x}_\Psi(t) \end{bmatrix} \\ &+ \begin{bmatrix} \mathcal{B}_\Delta \mathcal{N} & \mathcal{B}_w \\ \mathcal{B}_\Psi^{z\Delta} \mathcal{D}_\Delta \mathcal{N} + \mathcal{B}_\Psi^{w\Delta} \mathcal{N} & \mathcal{B}_\Psi^{z\Delta} \mathcal{D}_{\Delta w} \end{bmatrix} \begin{bmatrix} v_2(t) \\ w(t) \end{bmatrix}, \\ \begin{bmatrix} v_1(t) \\ z(t) \end{bmatrix} &= \begin{bmatrix} \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{C}_\Delta & \mathcal{C}_{\Psi,1} \\ \mathcal{C}_z & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_f(t) \\ \mathbf{x}_\Psi(t) \end{bmatrix} \\ &+ \begin{bmatrix} \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{D}_\Delta \mathcal{N} + \mathcal{D}_{\Psi,1}^{w\Delta} \mathcal{N} & \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{D}_{\Delta w} \\ \mathcal{D}_{z\Delta} \mathcal{N} & \mathcal{D}_{zw} \end{bmatrix} \begin{bmatrix} v_2(t) \\ w(t) \end{bmatrix} \end{aligned} \quad (27)$$

Say we want to control the given PIE system

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x}_f(t)) \\ z(t) \end{bmatrix} = \begin{bmatrix} \bar{\mathcal{A}}(\delta) & \bar{\mathcal{B}}_w(\delta) & \bar{\mathcal{B}}_u(\delta) \\ \bar{\mathcal{C}}(\delta) & \bar{\mathcal{D}}_{zw}(\delta) & \bar{\mathcal{D}}_{zu}(\delta) \end{bmatrix} \begin{bmatrix} \mathbf{x}_f(t) \\ w(t) \\ u(t) \end{bmatrix}$$

That admits the generalized plant $\{\mathcal{T}, (\mathcal{A} + \mathcal{B}_u \mathcal{K}), \mathcal{B}_\Delta, \mathcal{B}_w, (\mathcal{C}_\Delta + \mathcal{D}_{\Delta u} \mathcal{K}), \mathcal{D}_\Delta, \mathcal{D}_{\Delta w}, (\mathcal{C}_z + \mathcal{D}_{zu} \mathcal{K}), \mathcal{D}_{z\Delta}, \mathcal{D}_{zw}\}$, i.e.,

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x}_f(t)) \\ z_\Delta(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} + \mathcal{B}_u \mathcal{K} & \mathcal{B}_\Delta & \mathcal{B}_w \\ \mathcal{C}_\Delta + \mathcal{D}_{\Delta u} \mathcal{K} & \mathcal{D}_\Delta & \mathcal{D}_{\Delta w} \\ \mathcal{C}_z + \mathcal{D}_{zu} \mathcal{K} & \mathcal{D}_{z\Delta} & \mathcal{D}_{zw} \end{bmatrix} \begin{bmatrix} \mathbf{x}_f(t) \\ w_\Delta(t) \\ w(t) \end{bmatrix}$$

with,

$$w_\Delta(t) = (\Delta z_\Delta)(t).$$

The augmented input-output plant associated with Definition 15 then becomes

$$\begin{aligned} \hat{\mathcal{T}} &= \begin{bmatrix} \mathcal{T} & 0 \\ 0 & \mathcal{T}_\Psi \end{bmatrix}, \\ \hat{\mathcal{A}} &= \begin{bmatrix} \mathcal{A} & 0 \\ \mathcal{B}_\Psi^{z\Delta} \mathcal{C}_\Delta & \mathcal{A}_\Psi \end{bmatrix}, \\ \hat{\mathcal{K}} &= \begin{bmatrix} \mathcal{K} & 0 \\ 0 & 0 \end{bmatrix}, \\ \tilde{\mathcal{A}} &= \begin{bmatrix} \mathcal{B}_u & 0 \\ \mathcal{B}_\Psi^{z\Delta} \mathcal{D}_{\Delta u} & 0 \end{bmatrix}, \\ \hat{\mathcal{B}} &= \begin{bmatrix} \mathcal{B}_\Delta \mathcal{N} & \mathcal{B}_w \\ (\mathcal{B}_\Psi^{z\Delta} \mathcal{D}_\Delta + \mathcal{B}_\Psi^{w\Delta}) \mathcal{N} & \mathcal{B}_\Psi^{z\Delta} \mathcal{D}_{\Delta w} \end{bmatrix}, \\ \hat{\mathcal{C}} &= \begin{bmatrix} \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{C}_\Delta & \mathcal{C}_{\Psi,1} \\ \mathcal{C}_z & 0 \end{bmatrix}, \\ \tilde{\mathcal{C}} &= \begin{bmatrix} \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{D}_{\Delta u} & 0 \\ \mathcal{D}_{zu} & 0 \end{bmatrix}, \\ \hat{\mathcal{D}} &= \begin{bmatrix} (\mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{D}_\Delta + \mathcal{D}_{\Psi,1}^{w\Delta}) \mathcal{N} & \mathcal{D}_{\Psi,1}^{z\Delta} \mathcal{D}_{\Delta w} \\ \mathcal{D}_{z\Delta} \mathcal{N} & \mathcal{D}_{zw} \end{bmatrix}. \end{aligned}$$

such that

$$\begin{aligned} \partial_t(\hat{\mathcal{T}}\hat{\mathbf{x}}(t)) &= (\hat{\mathcal{A}} + \tilde{\mathcal{A}}\hat{\mathcal{K}})\hat{\mathbf{x}}(t) + \hat{\mathcal{B}}\hat{w}(t), \\ \hat{z}(t) &= (\hat{\mathcal{C}} + \tilde{\mathcal{C}}\hat{\mathcal{K}})\hat{\mathbf{x}}(t) + \hat{\mathcal{D}}\hat{w}(t) \end{aligned}$$

where $\hat{\mathbf{x}} = [\mathbf{x}_f(t)^* \quad \mathbf{x}_\Psi(t)^*]^*$, $\hat{z} = [v_1(t)^* \quad z(t)^*]^*$, $\begin{bmatrix} v_1^* & v_2^* \end{bmatrix}^* = \Psi \begin{bmatrix} z_\Delta^* & w_\Delta^* \end{bmatrix}^*$, and $\hat{w}(t) = [v_2(t)^* \quad w(t)^*]^*$. Then the dual system is

$$\begin{aligned} \hat{\mathcal{T}}^* \dot{\hat{\mathbf{x}}}(t) &= (\hat{\mathcal{A}}^* + \hat{\mathcal{K}}^* \tilde{\mathcal{A}}^*)\hat{\mathbf{x}}(t) \\ &+ (\hat{\mathcal{C}}^* + \hat{\mathcal{K}}^* \tilde{\mathcal{C}}^*)\hat{w}(t), \\ \hat{\hat{z}}(t) &= \hat{\mathcal{B}}^* \hat{\mathbf{x}}(t) + \hat{\mathcal{D}}^* \hat{w}(t) \end{aligned}$$

Moreover we also define

$$\tilde{\mathcal{V}} = \begin{bmatrix} \tilde{\mathcal{V}}_{11} & \tilde{\mathcal{V}}_{12} \\ \tilde{\mathcal{V}}_{21} & \tilde{\mathcal{V}}_{22} \end{bmatrix} = \begin{bmatrix} \mathcal{V}_{11} & 0 & \mathcal{V}_{12} & 0 \\ 0 & \hat{\mathcal{V}}_{11} & 0 & \hat{\mathcal{V}}_{12} \\ \mathcal{V}_{21} & 0 & \mathcal{V}_{22} & 0 \\ 0 & \hat{\mathcal{V}}_{21} & 0 & \hat{\mathcal{V}}_{22} \end{bmatrix} \quad (28)$$

Here the first and third block rows/columns of $\tilde{\mathcal{V}}$ correspond to the partition $v = [v_1^* \quad v_2^*]^*$. Since $w_\Delta = \mathcal{N}v_2$, the static multiplier is applied to the genuine output-input pair of the augmented map, and (28) acts on $\begin{bmatrix} v_1^* & z^* & v_2^* & w^* \end{bmatrix}^*$. If $\mathcal{V}$

Algorithm 1 Programming the Dual Synthesis LPI

Require: Generalized plant operators, basis degree ν , pole ρ , and multiplier templates for $\mathcal{V}_\Delta$ and $\tilde{\mathcal{V}}$.

- 1: Construct the basis filter and partition $v_1 = (\psi_\nu \otimes I_{z_\Delta})z_\Delta$, $v_2 = (\psi_\nu \otimes I_{w_\Delta})w_\Delta$.
- 2: Set $\mathcal{N} = \begin{bmatrix} I_{w_\Delta} & 0 & \cdots & 0 \end{bmatrix}$, so that $w_\Delta = \mathcal{N}v_2$.
- 3: Substitute $w_\Delta = \mathcal{N}v_2$ and form the augmented map $\hat{G} : \begin{bmatrix} v_2^* & w^* \end{bmatrix}^* \mapsto \begin{bmatrix} v_1^* & z^* \end{bmatrix}^*$.
- 4: Assemble the fixed operators $\hat{\mathcal{T}}, \hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}}, \hat{\mathcal{D}}, \hat{\mathcal{A}}, \hat{\mathcal{C}}$.
- 5: Declare decision variables $\mathcal{P} = \mathcal{P}^* \succ 0$, $\mathcal{Z} = \begin{bmatrix} \mathcal{Z}_{11} & \mathcal{Z}_{12} \\ 0 & 0 \end{bmatrix}$, and strict PN-PI multiplier variables.
- 6: Build $\tilde{\mathcal{V}}$ from (28) and compute its dual multiplier $\tilde{\mathcal{V}}'$.
- 7: Assemble the LPI in Theorem 4, using $\mathcal{Z} = \tilde{\mathcal{K}}\mathcal{P}$ instead of the controller $\tilde{\mathcal{K}}$.
- 8: Solve the LPI feasibility problem.
- 9: Recover the controller as $\mathcal{K} = \mathcal{Z}_{11}\mathcal{P}_{11}^{-1}$.

and $\hat{\mathcal{V}}$ are strict PN-PI operators, then $\tilde{\mathcal{V}}$ is also a strict PN-PI operator. If we would derive **robust performance** LPI without using the duality theorem, we would end up with a non-convex LPI, which we cannot efficiently solve using the PIETOOLS library. Therefore we apply the duality theorem to derive the following LPI.

Theorem 4 (Robust Controller Gain Synthesis). *Let G be parametrized by PI operators $\{\mathcal{T}, (\mathcal{A} + \mathcal{B}_u\mathcal{K}), \mathcal{B}_\Delta, \mathcal{B}_w, (\mathcal{C}_\Delta + \mathcal{D}_{\Delta u}\mathcal{K}), \mathcal{D}_\Delta, \mathcal{D}_{\Delta w}, (\mathcal{C}_z + \mathcal{D}_{zu}\mathcal{K}), \mathcal{D}_{z\Delta}, \mathcal{D}_{zw}\}$. Let $\Delta \in \Delta$, and there exists a strict PN-PI multiplier $\mathcal{M}_{\tilde{\mathcal{V}}}$ parameterized according to (28), such that*

- The interconnection $[G, \Delta]$ is well-posed for all $\Delta \in \Delta$.
- The IQC in Definition 9 is satisfied for all $\Delta \in \Delta$.
- The dynamic multiplier admits the partition $v = \begin{bmatrix} v_1^* & v_2^* \end{bmatrix}^*$ and a bounded selection operator $\mathcal{N}$ such that $w_\Delta(t) = \mathcal{N}v_2(t)$.
- The hatted operators $\{\hat{\mathcal{T}}, \hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}}, \hat{\mathcal{D}}\}$ denote the causal bounded realization $\hat{G} : \begin{bmatrix} v_2^* & w^* \end{bmatrix}^* \mapsto \begin{bmatrix} v_1^* & z^* \end{bmatrix}^*$ constructed above, and this realization and its dual scaled realization satisfy Theorem 3.

Then the interconnection $[G, \Delta]$ is **robustly stable**, with $\hat{x}(0) = 0$, and **robust performance** is achieved if there exist a PI operator $\mathcal{Z} : \mathcal{L}(\mathbb{R}^{n_*} \rightarrow \mathbb{R}^{2n_u})$, such that $\mathcal{Z} = \begin{bmatrix} \mathcal{Z}_{11} & \mathcal{Z}_{12} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \mathcal{K}\mathcal{P}_{11} & \mathcal{K}\mathcal{P}_{12} \\ 0 & 0 \end{bmatrix}$, a strict PN-PI multiplier $\mathcal{M}_{\tilde{\mathcal{V}}}$, with $\tilde{\mathcal{V}}_{11} \succeq 0$, and a coercive self-adjoint $\mathcal{P} \in \Pi_4$ with $\mathcal{P} = \mathcal{P}^* \succ 0$, such that,

- $\mathcal{K} = \mathcal{Z}_{11}(\mathcal{P}_{11})^{-1}$,

and

$$\begin{bmatrix} \hat{\mathcal{A}}\mathcal{P}\hat{\mathcal{T}}^* + \tilde{\mathcal{A}}\mathcal{Z}\hat{\mathcal{T}}^* + (*) & \hat{\mathcal{T}}\mathcal{P}\hat{\mathcal{C}}^* + \hat{\mathcal{T}}\mathcal{Z}^*\tilde{\mathcal{C}}^* + \hat{\mathcal{B}}\tilde{\mathcal{V}}'_{12} \\ \hat{\mathcal{C}}\mathcal{P}\hat{\mathcal{T}}^* + \tilde{\mathcal{C}}\mathcal{Z}\hat{\mathcal{T}}^* + \tilde{\mathcal{V}}'_{21}\hat{\mathcal{B}}^* & \tilde{\mathcal{V}}'_{21}\hat{\mathcal{D}}^* + \hat{\mathcal{D}}\tilde{\mathcal{V}}'_{12} + \tilde{\mathcal{V}}'_{22} + \epsilon I \\ \tilde{\mathcal{V}}'_{11}\hat{\mathcal{B}}^* & \tilde{\mathcal{V}}'_{11}\hat{\mathcal{D}}^* \end{bmatrix} \preceq 0$$

Here, $\tilde{\mathcal{V}}'$ is the dual multiplier of $\tilde{\mathcal{V}}$

Proof. Suppose the LPI holds. Define $V(\hat{x}) = \langle \hat{\mathcal{T}}^*\hat{x}, \mathcal{P}\hat{\mathcal{T}}^*\hat{x} \rangle \geq 0$. The LPI is derived in Appendix ?? . We

have that,

$$\dot{V}(\hat{x}(t)) + \epsilon \|\hat{w}(t)\|^2 + \left\langle \begin{bmatrix} \hat{z}(t) \\ \hat{w}(t) \end{bmatrix}, \tilde{\mathcal{V}}' \begin{bmatrix} \hat{z}(t) \\ \hat{w}(t) \end{bmatrix} \right\rangle_{\mathbb{R}L_2} \leq 0$$

Now, since $V(\hat{x}(0)) = V(0) = 0$ and $V(\hat{x}(T)) \geq 0$, after integrating in time, we obtain,

$$\int_0^T \left\langle \begin{bmatrix} \hat{z}(t) \\ \hat{w}(t) \end{bmatrix}, \tilde{\mathcal{V}}' \begin{bmatrix} \hat{z}(t) \\ \hat{w}(t) \end{bmatrix} \right\rangle_{\mathbb{R}L_2} dt \leq -\epsilon \int_0^T \|\hat{w}(t)\|^2 dt$$

Then by Theorem 3, Lemma 2 and Lemma 3, there exists $\epsilon > 0$ such that

$$\begin{aligned} & \int_0^T \left\langle \begin{bmatrix} v_1(t) \\ z(t) \\ v_2(t) \\ w(t) \end{bmatrix}, \tilde{\mathcal{V}} \begin{bmatrix} v_1(t) \\ z(t) \\ v_2(t) \\ w(t) \end{bmatrix} \right\rangle_{\mathbb{R}L_2} dt \\ & \leq -\epsilon (\|w_\Delta(t)\|_{\mathbb{R}L_2}^2 + \|w(t)\|_{\mathbb{R}}^2) \end{aligned}$$

Thus we can conclude

$$\begin{aligned} & \left\langle p_T \begin{bmatrix} v_1 \\ z \\ v_2 \\ w \end{bmatrix}, p_T \mathcal{M}_{\tilde{\mathcal{V}}} \begin{bmatrix} v_1 \\ z \\ v_2 \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\ & \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w\|_{L_2}^2), \end{aligned}$$

Therefore, by Theorem 1 in [4], robust performance and stability are guaranteed. $\blacksquare$

VII. NUMERICAL EXAMPLES

In this section, we use PIETOOLS [9] to construct the PIE representation and solve the associated LPIs for analysis. As a practical first step, it is generally advisable to synthesize a controller for the nominal system before accounting for uncertainties. In addition, we incorporate performance specifications through weighting filters. This is a standard technique in optimal and robust control synthesis for finite dimensional systems by embedding frequency-dependent performance characteristics into the synthesis.

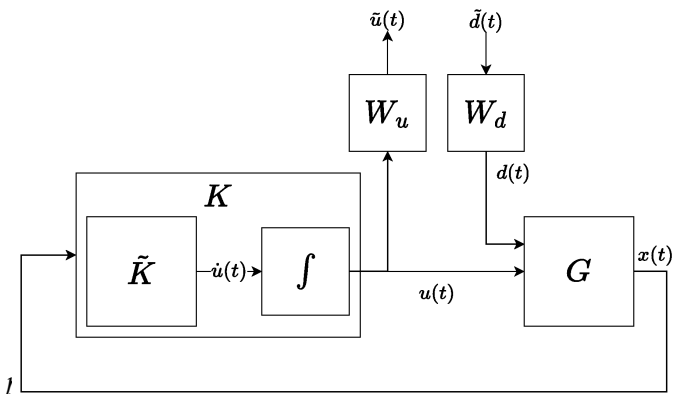


Fig. 2. Block schematic of the weighted diffusion-reaction system with boundary control, where W_u characterizes the inverse frequency-dependent characteristics of the desired control input $u(t)$, and W_d characterizes the frequency content of the disturbance, i.e., $\tilde{u}(t) = W_u u(t)$ and $w(t) = W_d \tilde{w}(t)$.

We begin by considering the boundary control problem for a diffusion-reaction equation, adapted from [8]. Throughout, we

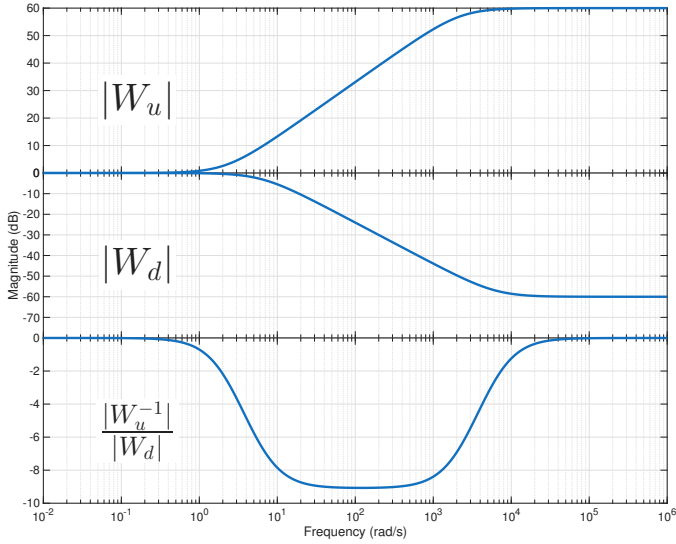


Fig. 3. Bode magnitude plots of the input and disturbance weighting filters, W_u and W_d , together with the combined inverse weighting $W_u^{-1}W_d^{-1}$. These weights are used to encode the desired frequency-dependent characteristics of the control input and disturbance in the weighted boundary control synthesis.

assume that the disturbance and control input are measurable, as the introduction of weighting filters gives rise to additional states that depend explicitly on these signals.

Example 1 (Reaction–Diffusion Equation). *Consider the boundary control of an unstable reaction–diffusion equation:*

$$\begin{aligned}\dot{\mathbf{v}}(t, s) &= 5\mathbf{v}(t, s) + \partial_s^2 \mathbf{v}(t, s) + C_{\tilde{d}} x_{\tilde{d}}(t) + D_{\tilde{d}} \tilde{d}(t), \\ \dot{x}_u(t) &= \dot{u}(t), \\ \dot{x}_{\tilde{u}}(t) &= A_{\tilde{u}} x_{\tilde{u}}(t) + B_{\tilde{u}} x_u(t), \\ \dot{x}_{\tilde{d}}(t) &= A_{\tilde{d}} x_{\tilde{d}}(t) + B_{\tilde{d}} \tilde{d}(t), \\ z(t) &= \tilde{u}(t) = C_{\tilde{u}} x_{\tilde{u}}(t) + D_{\tilde{u}} x_u(t), \\ \mathbf{v}(t, 0) &= 0, \quad \partial_s \mathbf{v}(t, 1) = x_u(t).\end{aligned}$$

Here, the state $x_{\tilde{u}}$ embeds the frequency characteristics of the weighting filter W_u , while $x_{\tilde{d}}$ embeds those of the weighting filter W_d shown in Fig. 3. Moreover, the input u is passed through an integrator so that the control input does not act directly on the boundary. This ensures that $\mathcal{T}_u = 0$, which is required for synthesis. Solving the associated synthesis conditions yields an $\mathcal{H}_\infty$ gain of $\gamma = 0.9773$, that is,

$$\frac{\|\tilde{u}\|_{L_2}}{\|\tilde{d}\|_{L_2}} < 0.9773.$$

This performance is achieved together with an exponential decay rate of $\alpha = 2$.

To assess whether the weighting specifications are met in practice, we compute an empirical transfer function estimate from the disturbance channel to the control input. For this purpose, we excite the system with a unit-amplitude frequency sweep ranging from 0.1 rad/s to 10^4 rad/s, filtered through the disturbance weighting filter W_d . The resulting signal is applied as the disturbance input, while the corresponding control input is measured. An estimate of the associated frequency response

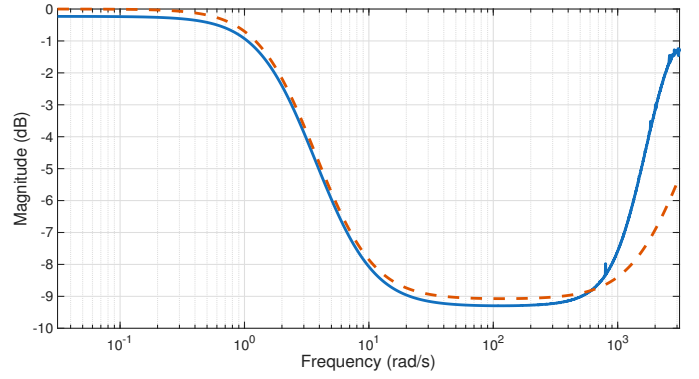


Fig. 4. Empirical frequency-response estimate from the disturbance input to the control input over the range 0.1 rad/s to 10^4 rad/s. — $|U|/|W|$ denotes the estimated transfer magnitude, while — $|W_u^{-1}|/|W_d|$ denotes the target weighting profile.

is then obtained from the Fourier transforms of these signals. Comparing this estimate with the desired weighted behavior allows us to verify that the imposed frequency-domain design specifications are achieved.

VIII. CONCLUSION

APPENDIX A MATHEMATICAL PROOFS

A. Proof of Theorem 1

Suppose that (7) holds. This implies,

$$\begin{aligned}& \left\langle p_T \Psi \begin{bmatrix} G_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_V \Psi \begin{bmatrix} G_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \\ & + \left\langle p_T \begin{bmatrix} G_{z\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\hat{V}} \begin{bmatrix} G_{z\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \\ & \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2\end{aligned}$$

Since $\hat{V}_{11} \succeq 0$, we can then infer,

$$\left\langle p_T \Psi \begin{bmatrix} G_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} G_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2$$

Then, using Theorem 6 in [11], we find that **robust stability** is guaranteed with respect to performance multiplier $\mathcal{M}_{\hat{V}}$. Moreover, rewriting (7) as,

$$\begin{aligned}& \left\langle p_T \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \\ & \left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\hat{V}} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w\|_{\mathbf{L}_{[a,b]}}^2),\end{aligned}$$

and given that Ψ and $\mathcal{M}_V$ satisfy the IQC in Definition 9 leads to,

$$\left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\hat{V}} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{L_2}^2.$$

Thus by Definition 10 **robust performance** is achieved.

B. Proof of Theorem 3

Sublemma 1. *The **Primal I/O Scaling** is a PI operator and invertible if and only if the **Dual I/O Scaling** is a PI operator and invertible*

Proof. To show sufficiency, let $\bar{\mathcal{S}} = (\hat{\mathcal{J}}^* \mathcal{S} \hat{\mathcal{J}})^{-*} = \hat{\mathcal{J}}^* \mathcal{S}^{-*} \hat{\mathcal{J}} = \hat{\mathcal{J}}^* (\mathcal{S}^{-1})^* \hat{\mathcal{J}}$, where $\hat{\mathcal{J}} = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} \in \Pi_4$.

Then if $\mathcal{S}$ is invertible then $\mathcal{S}^{-1} \in \Pi_4$ and by using that Π_4 forms a composition algebra we find that the **Dual I/O Scaling** is PI and invertible. For the necessary case, notice $\mathcal{S} = (\hat{\mathcal{J}}^* \bar{\mathcal{S}} \hat{\mathcal{J}})^{-*}$. ■

Sublemma 2. *If $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ and $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ are invertible, then the **Primal Scaled System** and **Dual Scaled System** are well-posed, respectively.*

Proof. Applying,

$$\begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ \mathcal{S}_{21} & \mathcal{S}_{22} \end{bmatrix} \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix},$$

on,

$$\begin{aligned} \partial_t(\mathcal{T}\mathbf{x})(t) &= \mathcal{A}\mathbf{x}(t) + \mathcal{B}w(t), \\ z(t) &= \mathcal{C}\mathbf{x}(t) + \mathcal{D}w(t). \end{aligned}$$

yields,

$$\begin{aligned} \partial_t(\mathcal{T}\mathbf{x})(t) &= \mathcal{A}\mathbf{x}(t) + \mathcal{B}(\mathcal{S}_{21}\tilde{z}(t) + \mathcal{S}_{22}\tilde{w}(t)), \\ (\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21})\tilde{z}(t) &= \mathcal{C}\mathbf{x}(t) + (\mathcal{D}\mathcal{S}_{22} - \mathcal{S}_{12})\tilde{w}(t). \end{aligned}$$

Then the **Primal Scaled System** is well-posed if and only if $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ is invertible, ensuring a unique solution for $\tilde{z}(t)$ in terms of $\mathbf{x}(t)$ and $\tilde{w}(t)$. A similar proof shows that the **Dual Scaled System** is well-posed if $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ is invertible. ■

To show sufficiency, let $\mathbf{x}(t)$, $w(t)$, and $z(t)$ satisfy the primal system dynamics with $\mathbf{x}(0) = 0$, and let $\tilde{z}, \tilde{w}$ be defined by the primal scaling in 1):

$$z = \mathcal{S}_{11}\tilde{z} + \mathcal{S}_{12}\tilde{w}, \quad w = \mathcal{S}_{21}\tilde{z} + \mathcal{S}_{22}\tilde{w}. \quad (29)$$

From the proof of Theorem 9 in [8], specifically (12), we established that the unscaled signals, $z(t)$, $w(t)$, $\tilde{z}(t)$ and $\tilde{w}(t)$, satisfy the identity:

$$\int_0^t \langle \tilde{z}(\theta), w(t-\theta) \rangle d\theta = \int_0^t \langle \tilde{w}(\theta), z(t-\theta) \rangle d\theta. \quad (30)$$

Substituting the dual scaling expressions from Eq. (29) into the left and right sides respectively, we obtain

$$\begin{aligned} & \int_0^t \langle \tilde{z}(\theta), \mathcal{S}_{21}\tilde{z}(t-\theta) + \mathcal{S}_{22}\tilde{w}(t-\theta) \rangle d\theta \\ &= \int_0^t \langle \tilde{w}(\theta), \mathcal{S}_{11}\tilde{z}(t-\theta) + \mathcal{S}_{12}\tilde{w}(t-\theta) \rangle d\theta. \end{aligned}$$

Rearranging the terms to group by $\tilde{z}$ and $\tilde{w}$ yields

$$\begin{aligned} & \int_0^t \langle \mathcal{S}_{22}^* \tilde{z}(\theta) - \mathcal{S}_{12}^* \tilde{w}(\theta), \tilde{w}(t-\theta) \rangle d\theta \\ &= \int_0^t \langle -\mathcal{S}_{21}^* \tilde{z}(\theta) + \mathcal{S}_{11}^* \tilde{w}(\theta), \tilde{z}(t-\theta) \rangle d\theta. \end{aligned}$$

We observe that the terms inside the inner product match the definition of the dual scaled variables:

$$\tilde{\tilde{w}} = -\mathcal{S}_{21}^* \tilde{z} + \mathcal{S}_{11}^* \tilde{w}, \quad \tilde{\tilde{z}} = \mathcal{S}_{22}^* \tilde{z} - \mathcal{S}_{12}^* \tilde{w}.$$

Substituting these identities, we arrive at the scaled identity:

$$\int_0^t \langle \tilde{\tilde{z}}(\theta), \tilde{w}(t-\theta) \rangle d\theta = \int_0^t \langle \tilde{\tilde{w}}(\theta), \tilde{z}(t-\theta) \rangle d\theta. \quad (31)$$

This equation is structurally identical to (30), but relates the scaled primal and dual variables. Now, we proceed with the truncation argument. For any fixed $\tau > 0$, let $\tilde{w}(t) \in \mathbb{R}_{L_2}$ be arbitrary. Under the assumption that the primal and dual scaled system are well-posed, i.e. $\mathcal{S}_{11} - \mathcal{D}\mathcal{S}_{21}$ and $\bar{\mathcal{S}}_{11} - \mathcal{D}^* \bar{\mathcal{S}}_{21}$ are invertible, the scaled primal input can be defined as $\tilde{w}(t) = \tilde{\tilde{z}}(\tau - t)$ for $t \leq \tau$ and $\tilde{w}(t) = 0$ for $t > \tau$. Then

$$\begin{aligned} \|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}}^2 &= \int_0^\tau \langle \tilde{\tilde{z}}(s), \tilde{w}(\tau - s) \rangle ds \\ &= \int_0^\tau \langle \tilde{\tilde{w}}(s), \tilde{z}(\tau - s) \rangle ds \\ &\leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}} \|P_\tau \tilde{z}(t)\|_{\mathbb{R}_{L_2}} \\ &\leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}} \|P_\tau \tilde{w}(t)\|_{\mathbb{R}_{L_2}}. \end{aligned}$$

Since by construction $\|P_\tau \tilde{w}(t)\|_{\mathbb{R}_{L_2}} = \|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}}$, thus we conclude

$$\|P_\tau \tilde{\tilde{z}}(t)\|_{\mathbb{R}_{L_2}} \leq \|P_\tau \tilde{\tilde{w}}(t)\|_{\mathbb{R}_{L_2}}$$

Since $\mathcal{T}^{**} = \mathcal{T}$, Sublemma 1, and

$$\begin{bmatrix} (\mathcal{A}^*)^* & (\mathcal{B}^*)^* \\ (\mathcal{C}^*)^* & (\mathcal{D}^*)^* \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix}^* = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}^{**} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}$$

we have that sufficiency implies necessity.

C. Proof of Lemma 2

Given the operators $\mathcal{Q}, \mathcal{S}, \mathcal{R} \in \Pi_4$

$$\mathcal{V} = \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix}.$$

Since $\mathcal{V}$ is a strict PN-PI operator, there exists $\epsilon > 0$ such that

$$\mathcal{Q} \geq \epsilon I, \quad \mathcal{R} \leq -\epsilon I.$$

In particular, $\mathcal{Q}$ is self-adjoint and coercive. Hence, by the positive square-root property for coercive self-adjoint PI operators, there exists a unique positive invertible operator $\mathcal{S}_{11}$ such that

$$\mathcal{Q} = \mathcal{S}_{11}^* \mathcal{S}_{11}, \quad \mathcal{S}_{12} := \mathcal{S}_{11}^* \mathcal{S}, \quad \mathcal{S}_{11}^* \mathcal{S}_{12} = \mathcal{S}.$$

Next define

$$\mathcal{W} := \mathcal{S}_{12}^* \mathcal{S}_{12} - \mathcal{R} = \mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} - \mathcal{R}.$$

Since $\mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} \geq 0$ and $-\mathcal{R} \geq \epsilon I$, it follows that

$$\mathcal{W} \geq \epsilon I.$$

Hence $\mathcal{W}$ is self-adjoint, coercive, and positive. Again by the positive square-root property, there exists a unique positive invertible operator $\mathcal{S}_{22}$ such that

$$\mathcal{W} = \mathcal{S}_{22}^* \mathcal{S}_{22}.$$

Therefore,

$$\mathcal{R} = \mathcal{S}_{12}^* \mathcal{S}_{12} - \mathcal{S}_{22}^* \mathcal{S}_{22}.$$

Combining these identities gives

$$\mathcal{V} = \begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ 0 & \mathcal{S}_{22} \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} \mathcal{S}_{11} & \mathcal{S}_{12} \\ 0 & \mathcal{S}_{22} \end{bmatrix}.$$

By construction, $\mathcal{S}_{11} > 0$ and $\mathcal{S}_{22} > 0$.

D. Proof of Lemma 3

Given the operators $\mathcal{Q}, \mathcal{S}, \mathcal{R} \in \Pi_4$, write

$$\mathcal{K} := \begin{bmatrix} -\mathcal{R} & \mathcal{S}^* \\ \mathcal{S} & -\mathcal{Q} \end{bmatrix}, \quad \bar{\mathcal{V}} = \mathcal{K}^{-1}.$$

Since $\mathcal{V}$ is a strict PN-PI operator, there exists $\epsilon > 0$ such that

$$\mathcal{Q} \succeq \epsilon I, \quad \mathcal{R} \preceq -\epsilon I.$$

so both $-\mathcal{R}$ and $\mathcal{Q}$ are self-adjoint, coercive, and invertible. Applying the block inverse formula to $\mathcal{K}$, generalized to the PI-operator setting from [5], gives the following expressions. In addition, [6, Theorem 2] shows that the inverse of a coercive, self-adjoint PI operator is again a coercive, self-adjoint PI operator.

$$\bar{\mathcal{Q}} = (\mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} - \mathcal{R})^{-1} \succeq \epsilon I,$$

and

$$\bar{\mathcal{R}} = -(\mathcal{Q} + \mathcal{S}(-\mathcal{R})^{-1} \mathcal{S}^*)^{-1} \preceq -\epsilon I.$$

Moreover, by composition algebra $\bar{\mathcal{S}}$ is also a PI operator,

$$\bar{\mathcal{S}} = \mathcal{Q}^{-1} \mathcal{S} \bar{\mathcal{Q}}$$

Hence $\bar{\mathcal{V}}$ is a strict PN-PI operator. The converse follows by the same argument with $\mathcal{V}$ and $\bar{\mathcal{V}}$ interchanged, since

$$\mathcal{V} = \begin{bmatrix} -\bar{\mathcal{R}} & \bar{\mathcal{S}}^* \\ \bar{\mathcal{S}} & -\bar{\mathcal{Q}} \end{bmatrix}^{-1}.$$

Thus $\mathcal{V}$ is a strict PN-PI operator if and only if $\bar{\mathcal{V}}$ is a strict PN-PI operator.

REFERENCES

- [1] A. Das, S. Shivakumar, M. Peet, and S. Weiland. Robust Analysis of Uncertain ODE-PDE Systems Using PI Multipliers, PIEs and LPIs. In *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 634–639, 2020.
- [2] Bin Hu, Márcio J. Lacerda, and Peter Seiler. Robustness analysis of uncertain discrete-time systems with dissipation inequalities and integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 27(11):1940–1962, 2017.
- [3] I. Köse and C. Scherer. Robust L_2 -gain feedforward control of uncertain systems using dynamic IQCs. *International Journal of Robust and Nonlinear Control*, 19(11):1224–1247, 2009.
- [4] Thijs Lenssen, Aleksandr Talitckii, Matthew Peet, and Amritam Das. A μ -Analysis and Synthesis Framework for Partial Integral Equations using IQCs, 2025.
- [5] T.-T. Lu and S.-H. Shiou. Inverses of 2×2 block matrices. *Computers & Mathematics with Applications*, 43(1):119–129, 2002.
- [6] Matthew M. Peet. A convex solution of the h_∞ -optimal controller synthesis problem for multi-delay systems, 2018.
- [7] Harald Pfifer and Peter Seiler. Less conservative robustness analysis of linear parameter varying systems using integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 26(16):3580–3594, 2016.

- [8] M. Peet S. Shivakumar, A. Das. Dual Representations and H_∞ -Optimal Control of Partial Differential Equations. *IEEE*, Volume(Issue):Start–End, 2026.
- [9] S. Shivakumar, A. Das, and M. Peet. PIETOOLS: A Matlab Toolbox for Manipulation and Optimization of Partial Integral Operators. In *2020 American Control Conference (ACC)*, pages 2667–2672, 2020.
- [10] S. Shivakumar, A. Das, S. Weiland, and M. Peet. Extension of the Partial Integral Equation Representation to GPDE Input–Output Systems. *IEEE Transactions on Automatic Control*, 70(5):3240–3255, 2025.
- [11] A. Talitckii, M. Peet, and P. Seiler. Integral Quadratic Constraints with Infinite-Dimensional Channels. In *2023 American Control Conference (ACC)*, pages 1576–1583, 2023.
- [12] Mi-Ching Tsai and Da-Wei Gu. *Robust and Optimal Control: A Two-Port Framework Approach*. Advances in Industrial Control. Springer, London, 2014.
- [13] Raghu Venkataraman and Peter Seiler. Convex LPV synthesis of estimators and feedforwards using duality and integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 28(3):953–975, 2018.